CO5 - AT1 - LEVEL-3

Level Exam   </> C   02 Sep 2026, 02:45 PM

← Back

Result Available

Your score: **71 / 100**

QUESTIONS

LEVEL 3 — HARD

Q1
function

Q2
functionsandpointers

2 / 2 saved

Level 3 — Hard **strong number (H)** — function

```
Write a function int isStrong(int n) that checks if a given number n is a strong number. A number is
considered strong if the sum of the factorial of its digits is equal to the number itself. The function should return 1 if the number
is strong, and 0 otherwise.
Sample Input:

Enter a number: 145

Sample Output:

145 is a strong number
```

Your Code

```c
#include <stdio.h>

int factorial(int num) {
    int fact = 1;
    for (int i = 1; i <= num; i++) {
        fact *= i;
    }
    return fact;
}

int isStrong(int n) {
    int original = n;
    int sum = 0;

```

Program Input (stdin)

Enter input if required...

Output

```
Enter a number: 145 is a strong number
```

Run   Save

CO5 - AT2 - Debugging

Level Exam   </> C   02 Sep 2026, 02:30 PM

← Back

Result Available

Your score: **90 / 100**

QUESTIONS

LEVEL 1 — EASY

Q1
Number

Q2
Number

Q3
Number

Q4
Number

4 / 4 saved

Level 1 — Easy **Q54** — Number

```
<pre>
Debug the following code:
#include<stdio.h>
int main(){
    int x = 10;
    printf("Value: %d\n", x);
}

</pre>
```

Your Code

```c
#include<stdio.h>
int main(){
    int x = 10;
    printf("Value: %d\n", x);
    return 0;
}
```

Program Input (stdin)

Enter input if required...

Output

```
Value: 10
```

Run   Save

QUESTIONS

Q1
CO5 - AT3 - HACKATHON / MINI PROJECT
100 marks

1/1 answered

Q1 CO5 - AT3 - HACKATHON / MINI PROJECT

100 marks

CO5 – AT3_HACKATHON QUESTIONS

1. A satellite continuously sends telemetry packets containing temperature, battery level, fuel status, altitude, and communication status to a ground station. Thousands of packets may arrive simultaneously from different processing channels. The system must store incoming records dynamically because the number of packets is unknown in advance.

Develop a C-based Real-Time Satellite Telemetry Processing System that:

• Uses dynamic memory allocation for telemetry records.
• Uses structures and pointers to manage satellite data.
• Uses functions to modularize packet processing.
• Uses POSIX threads to process multiple telemetry streams concurrently.
• Uses mutex/semaphore synchronization to prevent simultaneous updates to shared telemetry data.
• Stores processed telemetry in a file for later analysis.
• Uses command-line arguments to specify satellite ID and input/output files.
• Uses IPC or socket programming to simulate communication between the satellite simulator and ground station.

Hackathon Challenge: The system must continue processing telemetry even when multiple packets arrive at almost the same time without losing or corrupting records.

2. A hospital emergency department receives patient requests from multiple registration counters simultaneously. Each patient record contains patient ID, emergency level, symptoms, assigned doctor, and treatment status.

Develop a Smart Emergency Patient Coordination System in C that:

• Dynamically allocates memory for patients.
• Uses functions for registration, search, update, and discharge.
• Uses pointers to modify patient records.
• Uses threads to simulate multiple hospital departments.
• Uses mutexes to prevent two departments from updating the same patient simultaneously.
• Stores patient history in files.
• Uses command-line arguments to configure the hospital department.
• Uses IPC to transfer emergency information between registration and treatment modules.